A Common ASIP: Microcontroller

- For embedded control applications
  - Reading sensors, setting actuators
  - Mostly dealing with events (bits): data is present, but not in huge amounts
  - e.g., VCR, disk drive, digital camera (assuming SPP for image compression), washing machine, microwave oven
- Microcontroller features
  - On-chip peripherals
    - Timers, analog-digital converters, serial communication, etc.
    - Tightly integrated for programmer, typically part of register space
  - On-chip program and data memory
  - Direct programmer access to many of the chip's pins
  - Specialized instructions for bit-manipulation and other low-level operations

Trend: Even More Customized ASIPs

- In the past, microprocessors were acquired as chips
- Today, we increasingly acquire a processor as Intellectual Property (IP)
  - e.g., synthesizable VHDL model
- Opportunity to add a custom datapath hardware and a few custom instructions, or delete a few instructions
  - Can have significant performance, power and size impacts
  - Problem: need compiler/debugger for customized ASIP
- Remember, most development uses structured languages
- One solution: automatic compiler/debugger generation
  - e.g., www.tensillica.com
- Another solution: retargettable compilers
  - e.g., www.improvsys.com (customized VLIW architectures)

8051 Microcontroller

- Famous ASIP
- Developed by Intel in 1980
- Targeting embedded control
- Large variety of derivatives
  - With DSP
  - In FPGA
  - In ASIC

8051 features

- CPU
- RAM
- ROM
- I/O
- interrupt logic
- timer

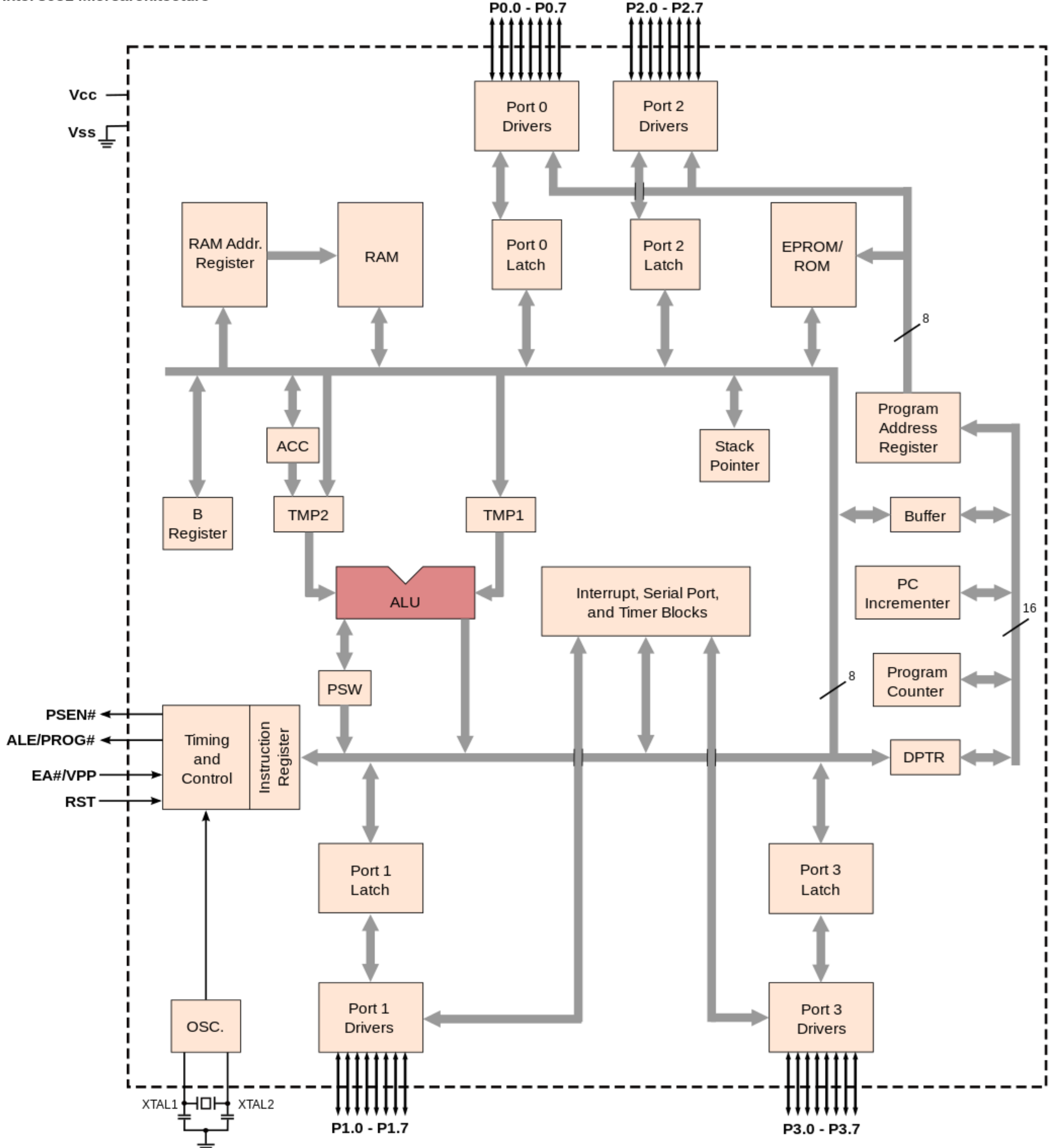
8051 features

- 8-bit ALU and accumulator
- 8-bit registers
- 8-bit data bus
- 2×16-bit address bus
- Four 8-bit bi-directional input/output port, bit addressable.
- 128 bytes of on-chip RAM
- 4 KB of on-chip ROM

8051 features

- 4 fast switchable register banks with 8 registers each (memory mapped)
- UART (serial port)
- Two 16-bit counter/timers
- Power saving mode (on some derivatives)

Intel 8051 Microarchitecture



Demo

- EdSim51 demo

8051 Assembly

- EdSim51 simulator: https://www.edsim51.com/
- Instruction set: http://www.keil.com/dd/docs/datashts/atmel/at_c51ism.pdf
- Useful website to quickly look at the details of an instruction:https://www.win.tue.nl/~aeb/comp/8051/set8051.html

Timing Notes

- 1 Clock period = 1/frequency of the oscillator
- 1 Machine cycle = 12 oscillator periods (In case of 8051, because of the inner structure)
- 1 Instruction cycle = 1-4 machine cycles (Depending on the instruction given for 8051)